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DRUG DISCOVERY TODAY DISEASE MECHANISMS

CNS and PNS diseases

Parkinson's disease – molecular mechanisms of disease

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It has long been recognized that exposure to environmental influences can increase the risk for developing idiopathic Parkinson's disease. The recent characterization of several novel gene mutations in mitochondrial and protein handling pathways has shed new light on the underlying cellular disturbances involved in familial Parkinson's disease. These findings point to cellular targets for drug development in Parkinson's disease; however, more studies are needed to identify potential biomarkers that can be used in clinical studies.

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Parkinson's disease is one of the most prevalent movement disorders of old age. Although the neuropathology of this disease has been well known for many years, the molecular mechanisms underlying neurodegeneration of substantia nigra and other monoaminergic cell types have only recently become better understood. Genetic studies in rare, familial forms of the disease have provided important clues to its underlying molecular causes. Thus, gene mutation in mitochondrial and protein handling pathways suggest that protein aggregation and reactive oxygen species are key contributions to pathology. Dr J.L. Eriksen and Dr L. Petrucelli have contributed greatly to our current understanding of molecular aspects of Parkinson's disease and they are thus well positioned to review the current status of this exciting field.

Introduction

Parkinson's disease (PD) is the most common neurologically based movement disorder, clinically diagnosed by the presence of bradykinesia, postural instability, resting tremor and rigidity [1]. This disease increasingly strikes with age, affecting about 1% of 65-year-old individuals and rising to 5% by 85 years of age. As a result of a progressive loss of dopaminergic neurons in the substantia nigra pars compacta (SNc), patients with PD experience an inexorable decline in their ability to complete even simple motor tasks; individuals gripped by advanced forms of the disease require full-time care. In the earliest stages of PD, symptoms are relatively well-controlled using levodopa and other dopamine-replacement therapies. However, the benefit of these agents is temporary, because these compounds do not impede the progression of the underlying neuropathology. Affected individuals become progressively less responsive to levodopa therapy, and can develop numerous neurological complications, including freezing, impaired autonomic function and dementia, caused by the degeneration of additional subcortical areas [2]. Given these observations, a major challenge in the development of PD therapy is to identify agents that can slow or reverse the progression of this disease by targeting the underlying neuropathological processes.

Main body

Targets for drug development

A crucial challenge involved in the generation of effective treatments for PD is to not only identify developing agents that can alleviate symptoms, but also to identify the appropriate pathogenic targets. In addition to the identification of novel genetic mutations, numerous studies have shown that environmental factors can exert influences on the development and progression of these diseases. Consequently, the current opinion of pathogenesis in sporadic PD favors a multifactorial process between genetic and environmental

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events. Important factors within this process include the formation of free radicals, impaired mitochondrial activity, increased sensitivity to apoptosis, excitoxicity and inflammation. Recent work suggests that impairment in the handling of misfolded and damaged proteins is another important contributor to pathogenesis [3]. These discoveries are beginning to yield new insights into appropriate targets for drug development.

Environmental risk factors

There is a substantial amount of data that supports the relationship between environmental agents and an increased incidence of PD. Epidemiological surveys have shown that the consumption of well water and degree of proximity to wood mills increase PD risk [2]. A significant association was found between the use of pesticides and herbicides in rural settings in the development of PD [4,5]. This finding has garnered additional support through experimental studies showing that exposure to certain pesticides and herbicides can cause the loss of dopaminergic neurons. Paraquat, a commonly used herbicide, accelerates α-synuclein fibril formation *in vitro*, increases α-synuclein levels in mice, and can cause irreversible nigrostriatal degeneration [6]. Rotenone, a pesticide, induces the formation of α-synuclein rich Lewy body-like inclusions and a loss of nigrostriatal neurons [7]. Compounds such as 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine (MPTP), a meperidine analogue first identified in street drugs, and annonacin, found in fruit and certain herbal teas, are linked to parkinsonism and can cause the degeneration of nigrostriatal pathways [2]. All of these toxins generate free radicals and act as direct inhibitors of the mitochondrial Complex I (NADH CoQ reductase enzyme; Fig. 1).

Work using the experimental toxin 6-hydroxydopamine (6-OHDA), a hydroxylated analogue of dopamine that is taken up through the dopamine transporter and selectively kills catecholaminergic neurons, has provided significant insight into the underlying mechanism of these neurotoxins. It has been firmly established that oxidative stress is a major precipitating factor of cell loss caused by 6-OHDA administration, involving the generation of hydrogen peroxidase and hydroxyl radicals in the presence of iron; furthermore, the neurotoxic effects can be blunted through the use of iron chelating compounds, monoamine oxidase B inhibitors (MAO-B) and antioxidants such as vitamin E that reduce the presence of free radicals [8]. However, although this compound kills dopaminergic neurons, it does not induce α-synuclein inclusions as is seen with rotenone treatment.

Two consistent biochemical 'signatures' in the substantial nigra of idiopathic PD cases are a deficiency of mitochondrial Complex I and abnormally high levels of free-radical damage. These two events are thought to be interrelated, because inhibition of Complex I activity can increase free radical production, and increased free radical production impairs Complex I activity. Studies that have attempted to modify the disease course in PD by reducing free-radical formation have shown relatively modest effects on PD. The Deprenyl and Tocopherol Antioxidative Therapy of Parkinsonism (DATATOP) study was a double-blinded, placebo-controlled trial using vitamin E as an antioxidant and deprenyl as a MAO-B inhibitor [9]. In addition to its antioxidant properties, it was thought that deprenyl would reduce free radical formation by reducing the turnover of dopamine. Although vitamin E was of no benefit in long-term treatment, deprenyl showed a modest protective effect. In a follow-up study of patients on levodopa, patients who also used deprenyl had a significant slowing of disease progression but increased levodopa-related side effects [10].

Coenzyme Q_{10} is another antioxidant with promise as a therapeutic agent in PD. This is an essential cofactor of the electron-transport chain of Complex I of the mitochondria and is a potent free radical scavenger in lipid and mitochondrial membranes. A small, double blinded study using coenzyme Q_{10} showed significant improvement in endpoint measures after 16 months of treatment [11]. However, before this agent is routinely recommended for patients diagnosed with PD, this small study must be replicated in larger trials.

Mitochondrial mutations

Recent surveys of individuals with familial Parkinson's disease have shown that mutations in three of the five known genes linked to PD are associated with mitochondrial impairment and decreased resistance to oxidative stress. The loss of *parkin* (GenBank accession number AB009973) is present in a rare familial, autosomal-recessive juvenile form of PD (ARJP) [12]. *Parkin* knockouts in fruit fly and mouse models show that these transgenics have impaired mitochondrial activity and a reduced resistance to oxidative stress. In *Drosophila*, muscle mitochondrial function appears to be compromised; in *parkin*-knockout transgenic mice, there is a reduction in mitochondrial and oxidative stress handling proteins, and an age-related increase in oxidatively damaged proteins.

Two other PD genes were recently identified that are associated with impaired mitochondrial function. In a rare, recessive form of PD, Bonifati *et al.* [13] discovered two different alterations in *DJ-1* (GenBank accession number AB073864), both resulting in a loss of protein function. Mutations in *PINK1* (PTEN-induced kinase; GenBank accession number AB053323), a putative mitochondrial gene, have been identified in an autosomal recessive, early onset form of PD [14]. Although the functions of these proteins are unknown, both DJ-1 and PINK1 co-localize with mitochondria, and the mutant forms of these proteins sensitize cells to oxidative stress. It has not been determined whether altered expression of these genes is significant in sporadic forms of the disease.

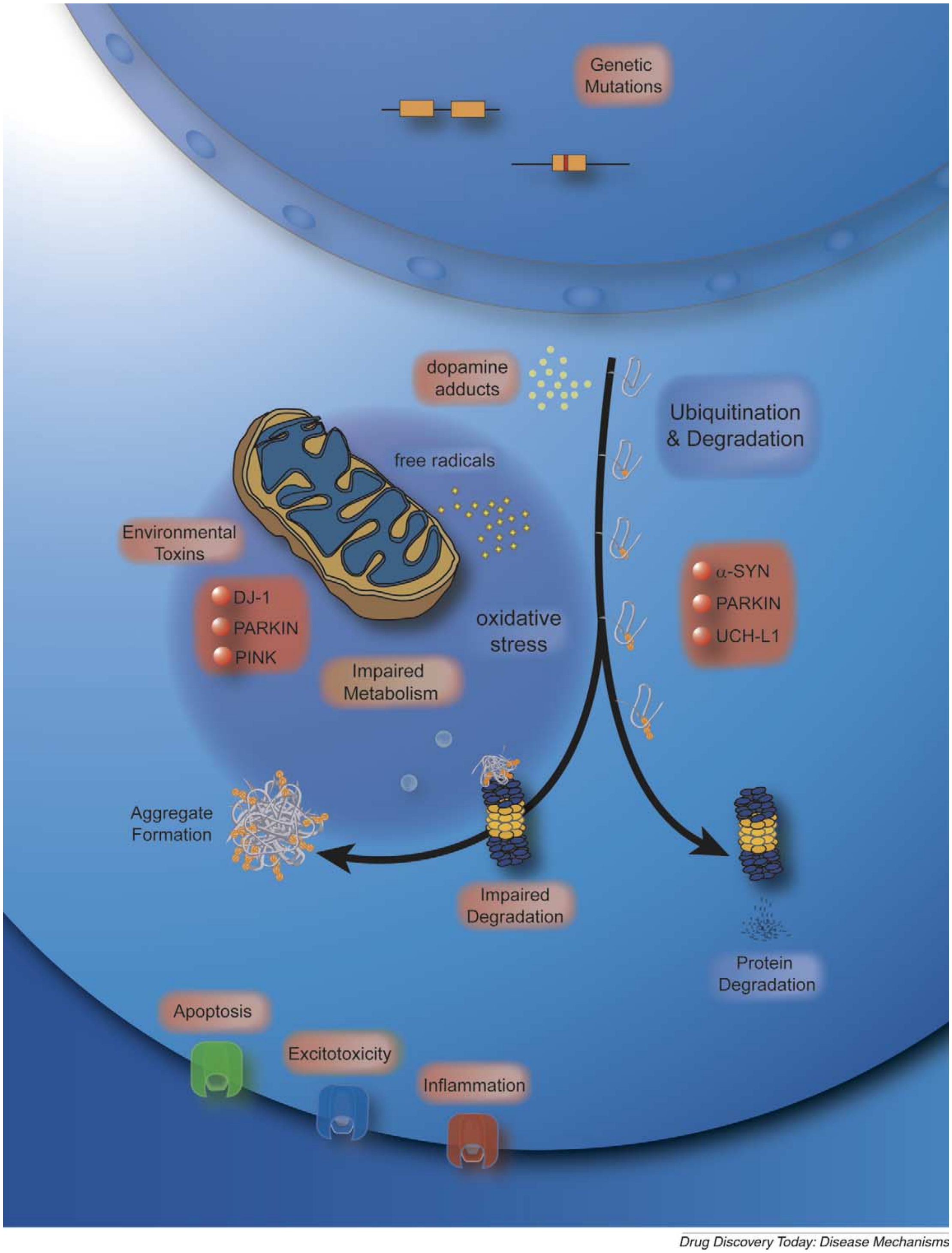

Figure 1. Potential drug targets in PD. Mutations in five genes are known to cause early-onset PD, including *α-synuclein*, *UCHL1*, *parkin*, *DJ-1*, and *PINK1*. Mutations and altered expression of in these genes can contribute to PD pathogenesis through common mechanisms related to proteasome inhibition, mitochondrial impairment and increased oxidative stress. Excessive intracellular accumulation of α-synuclein, resulting in spontaneous misfolding and oligomerization of this protein, can lead to the formation of toxic protein. Dopamine can also form toxic adducts with α-synuclein. Mutant α-synuclein can inhibit proteasomal activity, potentially leading to enhanced accumulation; soluble α-synuclein protofibrils and intracellular protein aggregates have been shown to impair the proteasome. Mutations and post-translational modifications of parkin and UCHL1 are associated with the UPS pathway and disruption of these proteins might increase proteotoxic stress. Loss of parkin function is thought to result in mitochondrial impairment. DJ-1 and PINK1 protect against oxidative stress and co-localize with the mitochondria. Disruption of these proteins might decrease mitochondrial activity, potentially impairing proteasomal function. Common environmental toxins, such as paraquat and rotenone, impair mitochondrial dysfunction, generate free radicals and increase oxidative stress, potentially resulting in proteasome dysfunction that can lead to the aggregation of α-synuclein. Combined with other cellular stress, this leads to neuronal loss.

The accumulation of unwanted proteins in PD

Lewy bodies are a major pathological hallmark of PD. These protein structures are intracytoplasmic aggregates composed primarily of α-synuclein and ubiquitin, along with smaller quantities of other proteins [1]. Mutations in α-synuclein appear to increase the levels of this protein, implying a toxic 'gain-of-function.' In pedigree studies of early onset, familial PD, the first mutation linked to this disease was found in the α-synuclein gene (GenBank accession number NM007308), and subsequent linkage analysis studies have identified several additional point mutations [15,16]; these mutations can promote the rapid aggregation of these proteins. More recently, the discovery of a genomic triplication of α-synuclein in a large, early-onset PD family demonstrated that the increased levels of wild-type α-synuclein might be the precipitating cause of pathogenesis in sporadic PD [17].

Experimental models using of α-synuclein have replicated much of the pathology found in humans. In transgenic mice, overexpression of wild-type and mutant forms of human α-synuclein resulted in the loss of dopaminergic terminals, the formation of α-synuclein rich inclusions and the development of motor deficits in mature animals [18]. Similar effects were found with mutant and wild-type forms of human α-synuclein in fruit flies [19]. A knockout of α-synuclein in mice had no detrimental effects on these animals, suggesting that overexpression of this protein leads to a toxic gain of function that leads to Lewy body formation [20]; conversely, synuclein knockout mice appear to be protected against MPTP-toxicity. It is not clear which forms of α-synuclein (protofibrils or Lewy bodies) exert the majority of toxic effects in the pathogenic PD cascade. Protofibriliar α-synuclein could be the toxic species because pathogenic mutations promote the formation of soluble protofibrils [21]. Studies of protein aggregates are more equivocal. Cell-based studies show that protein aggregates directly impair the ubiquitin–proteasome system (UPS) [22]; other studies suggest that aggregation is a protective effect. Regardless of the ultimate toxic form of the protein, strategies that limit α-synuclein accumulation are likely to be useful in PD therapeutics.

Protein mishandling in PD

Under normal circumstances, most toxic and damaged proteins are degraded through the ubiquitin–proteasome system. Proteins destined for degradation are bound with chaperones (proteins that recognize misfolded/abnormal proteins) and tagged with ubiquitin moieties. Ubiquitination is a highly ordered and ATP-dependent process, where lysine residues on the target molecule are tagged with ubiquitin moieties in an three-step process (denoted E1–E3). The E2 and E3 ligases add multiple ubiquitins, forming a chain that allows tagged proteins to undergo proteolytic degradation through the 26S proteasome [23].

The hypothesis that protein mishandling might be of some importance (Fig. 2) gained considerable momentum with the discovery that a loss of the *parkin* gene was associated with autosomal recessive PD (in ARJP) [12]. Parkin is an E3 ligase that ubiquitinates proteins for proteasomal degradation, and it was believed that the loss of this protein might be responsible for dopaminergic cell loss in ARJP. However, studies using fruit fly and transgenic mouse models found that deletion of this gene caused general cell loss, rather than targeted dopaminergic cell loss, and subtle alterations in mitochondrial and oxidative stress proteins [24–26].

Subsequent to the discovery of *parkin*, mutations in ubiquitin carboxyhydrolase L1, a deubiquitating enzyme (GenBank accession number CM981986, UCH-L1), were associated with a rare form of early-onset PD, demonstrating that impairment in UPS function might play a role in the development of this disease [27]. Biochemical studies have also underscored this association. For instance, the analysis of striatal samples from PD patients show a loss of UPS degradation activity [28]. Cell-based studies have also demonstrated that mutant forms of α-synuclein directly impair UPS activity in catcholamenergic cells, offering one explanation as to why dopamine neurons are particularly sensitive to loss in PD [29]. Finally, a new gene responsible for the PARK8 parkinsonian phenotype, dubbed *dardarin*, appears to have a kinase domain that could modulate potentially modulate protein degradation through it phosphorylation activity, although the function of this gene is currently unknown [30].

Some of the most compelling *in vivo* work supporting an association between PD and the UPS comes from a recent report that two-week, partial inhibition of the UPS system in rodents leads to severe, irreversible PD-like symptoms [31]. Rats given proteasomal inhibitors showed behavioral signs that were remarkably similar to PD patients, including bradykinesia, rigidity and resting tremor. These symptoms progressed over several months and improved with levodopa treatment. In addition to dopaminergic cell loss, the animals developed Lewy body-like inclusions in brain regions affected in PD.

Although there are no strategies currently available that directly activate UPS activity, Hsp90 inhibitors such as geldanamycin are known to enhance protein degradation. Several groups have focused on overexpression of heat shock proteins (HSPs) to sequester and degrade pathogenic α-synuclein species. *In vivo* studies in fruit fly and transgenic α-synuclein mice have been encouraging. The overexpression of Hsp70 (the major inducible HSP), or treatment with geldanamycin, prevented neurodegeneration normally seen in the *Drosophila* PD model [32,33], and reduced the insoluble α-synuclein species formed in the transgenic mouse line [34]. Although additional experimental studies are needed, this data support the hypothesis that overexpression of

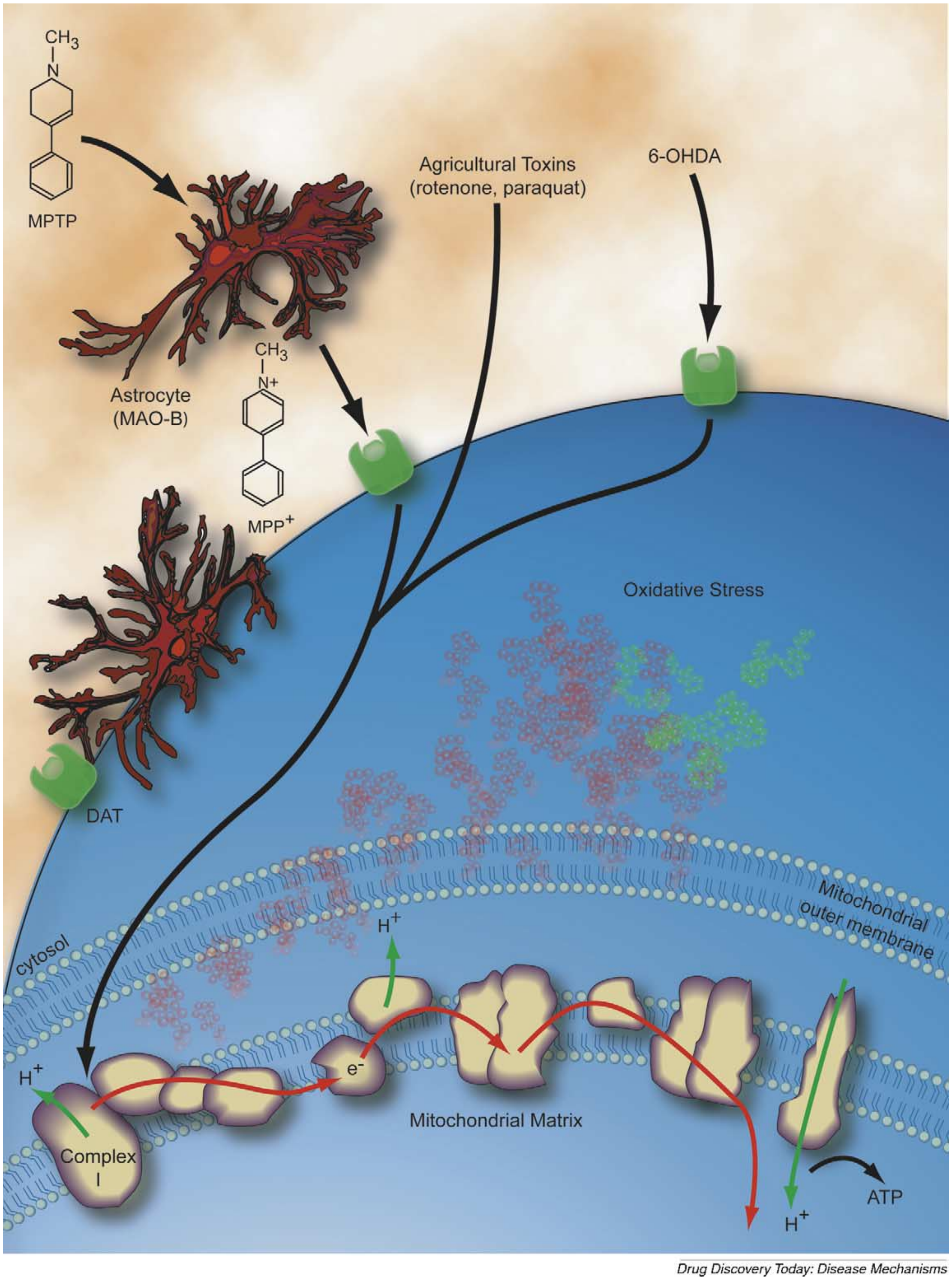

Figure 2. Intracellular pathways of dopaminergic neurotoxins in animal models of Parkinson's disease. MPTP is taken up by astrocytes and catalyzed by enzyme monoamine oxidase-B, *MAO-B*, into MPP^+ (its active form). MPP^+ can be specifically transported into dopaminergic neurons via the dopamine transporter (DAT). Similarly, the hydroxylated analogue of dopamine, 6-OHDA, is selectively taken up via DAT by dopaminergic neurons. Agricultural toxins, such as rotenone and paraquat, are able to freely enter the dopaminergic neuron and accumulate inside mitochondria. All of these compounds target Complex I of the mitochondrial electron-transfer chain (ETC), consists of Complex I–V (*C-I*, *C-II*, *C-III*, *C-IV*, *C-V*), resulting in the generation of reactive oxygen species (ROS); free radicals can interact with iron present in the cell, exacerbating neurotoxicity. All of these mitochondrial toxins enhance the production of free radicals and decrease the synthesis of ATP.

Table 1. Therapeutic agents in Parkinson's disease

Target	Agent	Results/trial	Refs
Apoptosis	CEP 1346[a]	FDA Phase III (Lundbeck)	[2]
	Minocycline	Possibly protective	[35,36]
	Rasagiline	Protective	[2]
	Deprenyl (Selegiline)	DATATOP. Modest protection	[9]
	TCH 346[b]	FDA Phase I (Novartis)	[2]
Bioenergetics	Coenzyme Q10	Delayed PD progression	[11]
Degradation pathways	Hsp90 inhibitors	No current data	[32,33]
Excitotoxicity	Remacemide	No effect	[2]
	Riluzole	No effect; trials halted	[2]
Inflammation	Minocycline	Experimentally protective	[35,36]
	NSAIDs[c]	Reduced risk by 45%	[39]
Oxidative stress	Coenzyme Q10	Delayed PD progression	[11]
	Rasagiline	Protective	[2]
	Deprenyl (Selegiline)	DATATOP. Modest protection	[9]
	Vitamin E	No effect	[9]

This table shows some compounds tested in Parkinson's disease.
[a] CEP 1346 inhibits a mixed lineage kinase family (MLK) that regulates the c-jun N-terminal kinase pathway.
[b] TCH 346 inhibits glyceraldehyde-3-phosphatase dehydrogenase activity.
[c] NSAIDs, non-steroidal anti-inflammatory drugs.

chaperones is likely to be a useful molecular target in the treatment of PD.

Other pathogenic pathways

There are several other therapeutic targets of interest in PD research, including apoptotic cell death, active inflammatory processes and excitotoxic events. However, their direct pathogenic significance is not clearly defined in PD, and clinical trials that have tested agents designed to halt these toxic processes have been inconclusive (see Table 1). Apoptosis (programmed cell death) has been shown to be a potential pathway of cell loss in PD. The administration of minocycline, a tetracycline derivative that inhibits caspases and apoptosis, partially protected against the loss of striatal dopaminergic neurons [35] in MPTP-treated mice, although contradictory results showing an exacerbation of toxicity were recently reported [36].

Minocycline was also found to inhibit microglial activation and release of potentially toxic cytokines caused by inflammatory processes in the striatial region of MPTP mice, which might explain some of the protective effects. In PD patients, there is substantial microglial activation and increased expression of pro-inflammatory cytokines [37,38]. A prospective study that investigated the relationship between use of common non-steroidal anti-inflammatory agents (NSAIDS) and PD risk showed that patients taking non-aspirin NSAIDS lowered their potential risk by 45% [39]. There is also some evidence that excitotoxic damage in the substantial nigra occurs in PD, but the importance of this to PD pathogenesis is unknown. Two large placebo-controlled studies using riluzole, a weak glutamate antagonist, were stopped owing to a lack of effect, but more potent glutamatergic antagonists could have been more effective [2].

Conclusion

Recent discoveries in early-onset PD have begun to provide new experimental targets for the development of therapeutic agents. Although there appears to be many potential driving forces behind the pathogenesis of this disease, the majority of mutations associated with early-onset PD indicate that there are disturbances in the relationship between mitochondrial and UPS activity, and therefore it can be argued that disruption of either system results in the development of PD.

Degradation of proteins through the UPS is highly dependent upon ATP generated from mitochondrial metabolism. In experimental systems, rodents treated with environmental toxins such as rotenone or MPTP develop PD-like behavioral symptoms, loss of dopaminergic cells in the substantial nigra and Lewy body-like inclusions. Similarly, the short-term

Links

- National Parkinson Foundation (http://www.parkinson.org/)
- CenterWatch.org (http://www.centerwatch.org/)
- Michael J. Fox Foundation (http://www.michaeljfox.org/)
- Information on Clinical Trials and Human Research Studies (http://www.clinicaltrials.gov/)
- MIT OpenCourseWare: Biological Chemistry II – Protein folding and degradation (http://ocw.mit.edu/OcwWeb/Chemistry/5-08Spring2004/)
- National Institute of Neurological Disorders and Stroke (NINDS) (http://www.ninds.nih.gov/)

administration of proteasomal inhibitors in rats results in the clinical and pathologic phenotypes observed in PD.

Given these findings, it is tempting to speculate that UPS activity is the rate-limiting factor in the development of PD, and that an important molecular target is the pathway that regulates α-synuclein levels. This hypothesis is bolstered by the recent report of the wild type α-synuclein triplication found in a large early-onset familial form of PD; the increased level of this protein is all that is required to develop the disease. Nevertheless, there is a significant quantity of evidence that supports contributions from other pathogenic pathways, such as the results showing NSAIDs and antioxidant treatment delays disease. The discovery of appropriate molecular targets and associated biomarkers in PD remains a work in process, but recent developments in the field have demonstrated that the immediate future holds much promise for the development of effective PD agents.

Outstanding issues

- Clinical protocols and genetic testing for earlier and more precise clinical diagnosis.
- To define the molecular targets involved in PD.
- Appropriate biomarkers for disease stage.
- Need for expanded clinical studies into areas with experimentally promising results.
- Relationships between energy metabolism and protein degradation pathways in PD.

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